

KUNAL JADHAV

Robotics Engineer · ROS2 · Computer Vision · Autonomous Systems
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EDUCATION

Lovely Professional University

Bachelor of Technology - Mechanical Engineering (Mechatronics)

Relevant Courses: Control Systems, Mechatronics System Modelling, Industrial Automation, Microprocessors & Microcontrollers, Electrical Machines, DFMEA, Programming

Jalandhar, Punjab, India

Aug 2020 – Jul 2024

TECHNICAL SKILLS

Robotics & Middleware: ROS2 (Humble), Nav2, Gazebo, RViz, URDF, TF2, RQT

Computer Vision & AI: OpenCV, Media Pipe, YOLO, Image Processing, Gesture Recognition

Programming: Python (NumPy, Matplotlib), C++, Bash/Shell scripting

Tools & Hardware: Linux (Ubuntu), Git, Docker, Raspberry Pi, Arduino, SolidWorks, Creo Parametric, PLC, SCADA (ABB)

PROJECTS

Vision-Driven Semantic & Gesture Control with Nav2 | ROS2 · Nav2 · OpenCV · Media Pipe · Python · Gazebo

Developed an integrated perception navigation system combining semantic scene understanding with real-time hand-gesture control over Nav2. The pipeline fuses camera input with Media Pipe hand-landmark detection to issue high-level navigation goals, while semantic segmentation classifies the environment to enable context-aware path planning and dynamic obstacle re-routing.

Autonomous Navigation Drone | ROS2 · Nav2 · Gazebo · URDF · Python · XML

Architected a full drone navigation system using ROS2 in Gazebo for safe validation of dynamic path planning and obstacle avoidance algorithms before physical deployment.

Terrabot — Four-Wheeled ROS2 Explorer | ROS2 Humble · Gazebo Fortress · Python · URDF · Linux

Developed a modular four-wheeled robot simulation using the ROS-Gazebo bridge, implementing real-time velocity control with synchronized physics and custom motion nodes for robust sensor integration.

Turtlebot3 Gazebo Simulation | ROS2 · Nav2 · Python · Linux

Built a complete Turtlebot3 simulation environment with ROS2 and Nav2, designing and testing custom nodes for motion control, real-time LiDAR integration, and autonomous waypoint navigation.

Air Canvas — Touchless Computer Vision Interface | Python · Media Pipe · OpenCV

Built a touchless drawing application using Media Pipe hand-landmark detection for seamless mode switching under variable lighting. demonstrates robust vision pipeline design with real-time performance constraints.

Customized Automated Wheelchair Sanitation System | Raspberry Pi · Python · ROS2 · SolidWorks

Invented and patented a cost-effective wheelchair sanitation system, leading the full innovation lifecycle from research and SolidWorks CAD design through embedded Python and ROS2 prototyping for improved patient hygiene.

ROS2 Virtual Object Finder with OpenCV | Arduino UNO · SolidWorks · Rapid Prototyping

Architected and deployed a virtual ROS2 robotic node equipped with OpenCV-driven visual processing, enabling real-time object detection, environment scanning and dynamic navigation within simulated environments.

Industrialized Robotic Arm with Bluetooth Control and Arduino Module | Arduino UNO · SolidWorks · Rapid Prototyping

Led the mechanical design and structural assembly of an automated robotic arm, collaborating on the embedded Arduino programming and wireless HC-05 Bluetooth integration to deliver a fully controllable prototype for industrial applications.

PATENTS & PUBLICATIONS

- Modular Self-Dismantling Drone for Multi-Point Surveillance** — Indian Patent Application No. 202511060537 (Filed May 2025)
- Customized Automated Wheelchair Sanitation System** — Indian Patent No. 202311078752 A (Issued Dec 2023)

LEADERSHIP & VOLUNTEERING

Pahal Student Organization, LPU — Director

Aug 2020 – Mar 2023

- Led a team of 110+ student volunteers organizing technical workshops and social welfare drives, building collaborative project management skills directly applicable to research group environments.

National Service Scheme, LPU — Volunteer

Mar 2023 – Feb 2024

- Orchestrated large-scale community service events managing on-ground logistics and cross-functional volunteer coordination for 100+ participants.

RELEVANT CERTIFICATIONS

- ROS2 Beginner on UDEMY [Certificate](#)
- OpenCV and Image Processing via Coursera [Certificate](#)
- Forage Robotics Engineer Job Simulation [Certificate](#)